

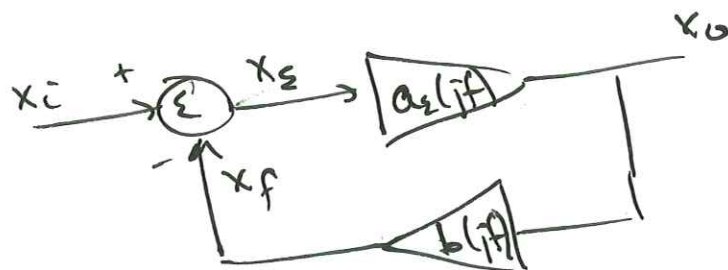
Dynamic Operation of Op Amps & Op Amp Circuits

$$a \rightarrow a(jf)$$

$$a_e \rightarrow a_e(jf)$$

$$b \rightarrow b(jf)$$

$$\rightarrow \text{L} = a_e b, \rightarrow L(jf) \rightarrow T(jf)$$



Need negative feedback for
amp circuit to work
linearly

~~a~~ $L(jf) \rightarrow$ now a complex #, stability more uncertain (..?)

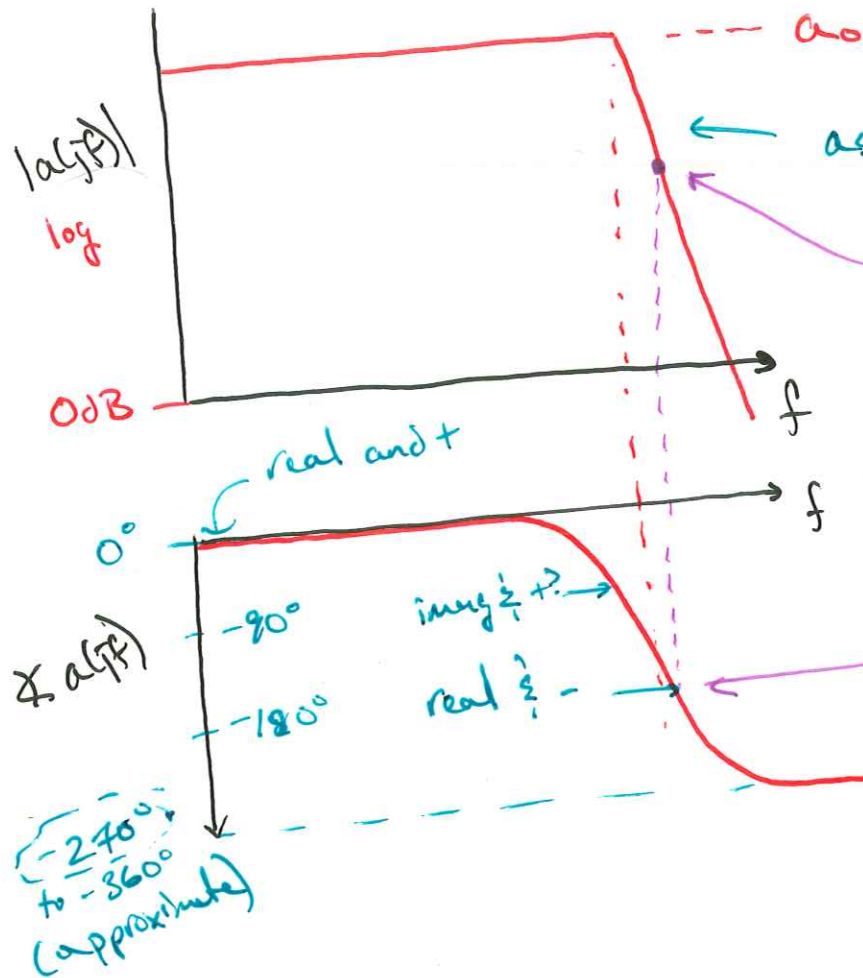
Consider Non-Inv. Amp...

amplifier is still $a(jf)$ but feedback is
completely real (assumed).

$$\therefore b(jf) \rightarrow b$$

Generic Uncompensated Amplifier (op Amp)

a_0 = open loop gain of amp at DC



assume several poles in response

At Positive feedback, gain is $> 0dB = 1/V$

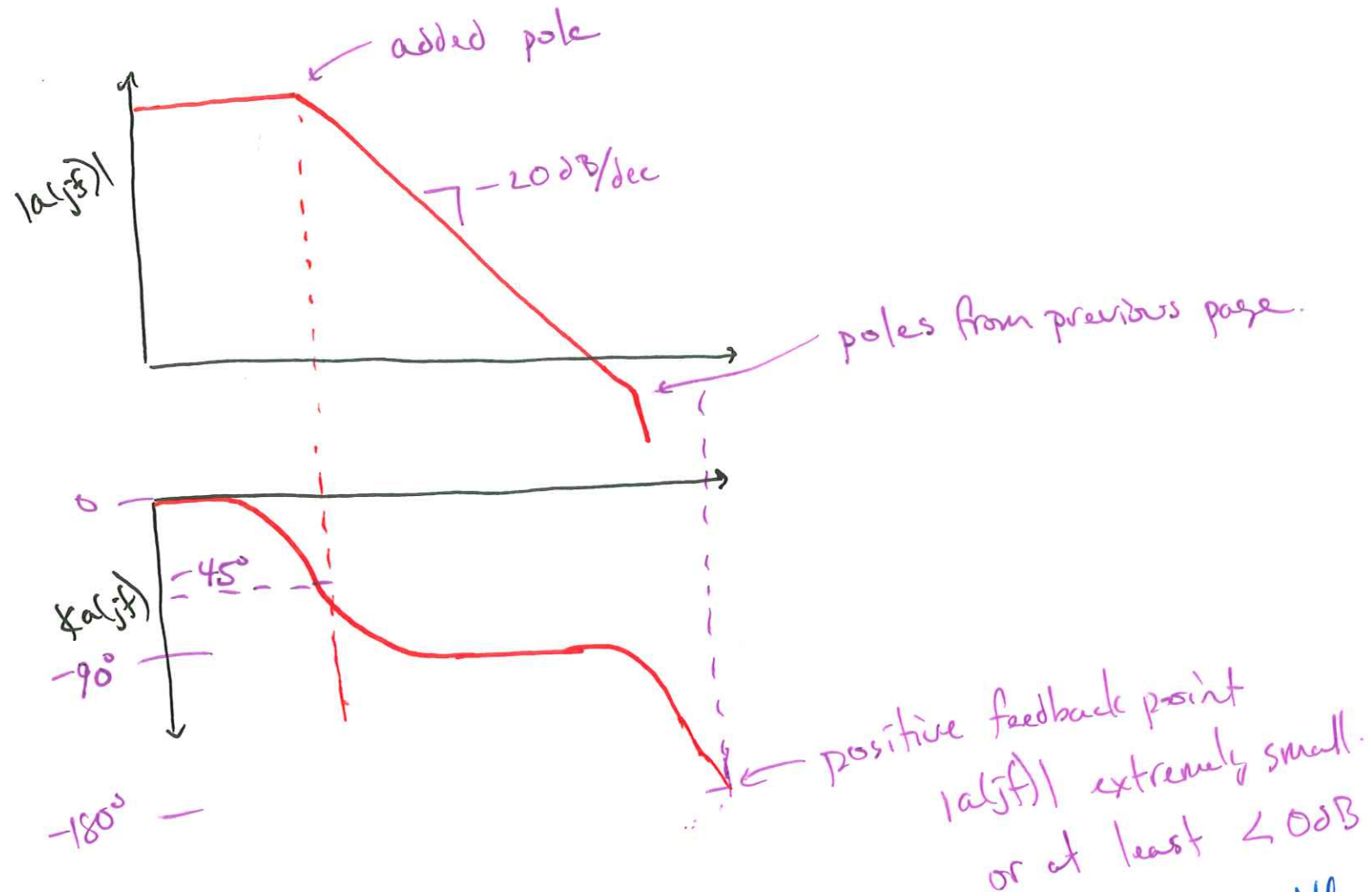
Loop gain Real is Negative $\rightarrow$ Positive Feedback!

unstable!

$\rightarrow$ to avoid instability, need $|L(jf)| \approx |T(jf)| < 0dB$ at $\angle L(jf), \angle T(jf) = -180^\circ$

Compensated Op Amps

Add pole to reduce $|a(jf)|$ when $\angle a(jf) = -180^\circ$



Significant loss of open loop Bandwidth
but now stable.

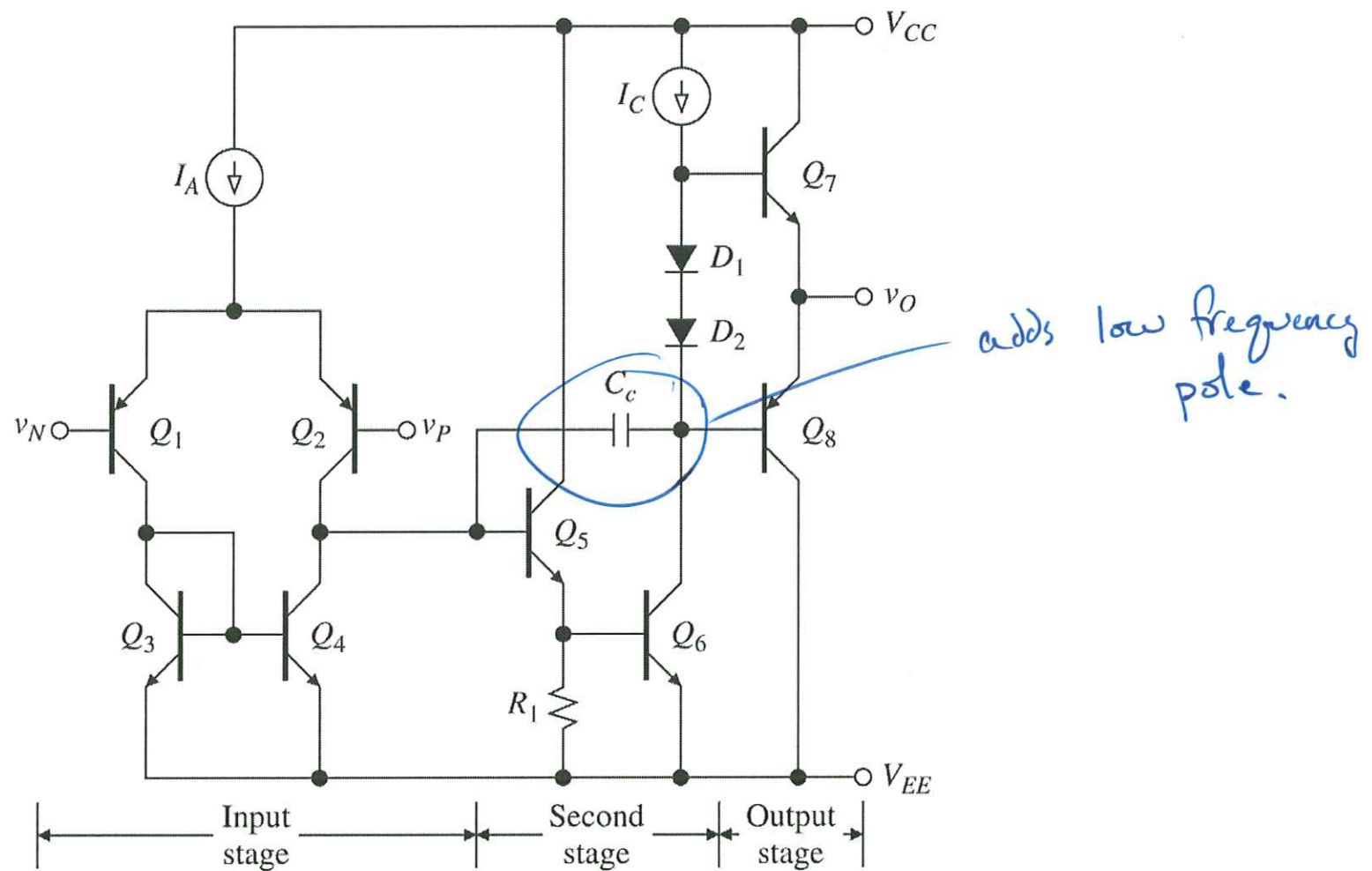
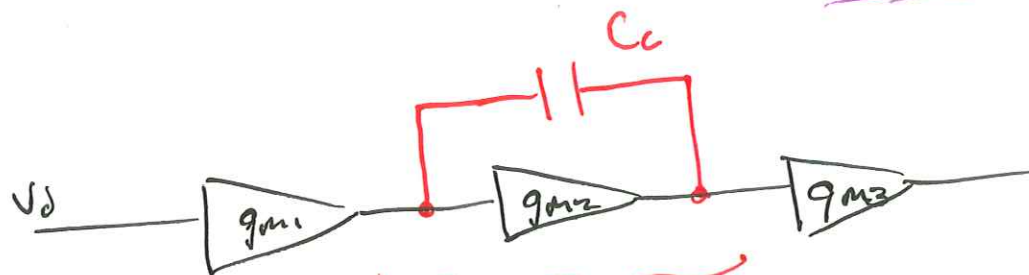


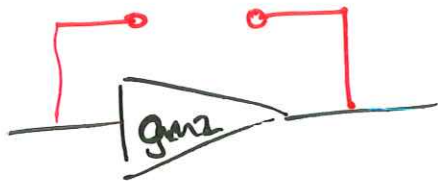
Figure 5.1: Simplified circuit diagram of a typical bipolar op amp.

Adding low-frequency Pole:

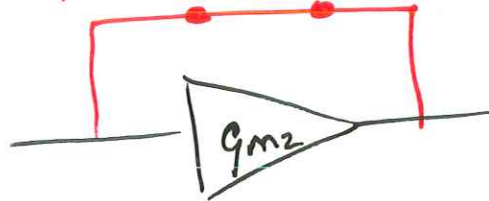
Dominant Pole Op-Amp



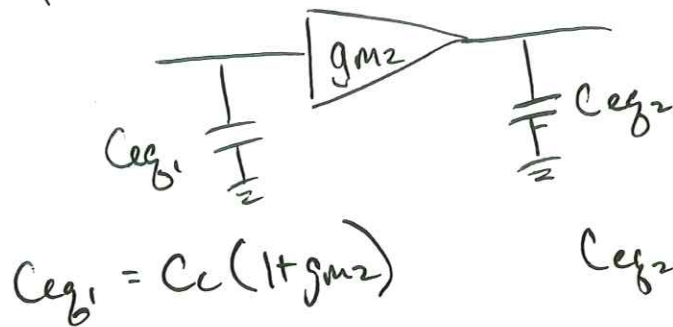
low f



high f



By Miller's theorem



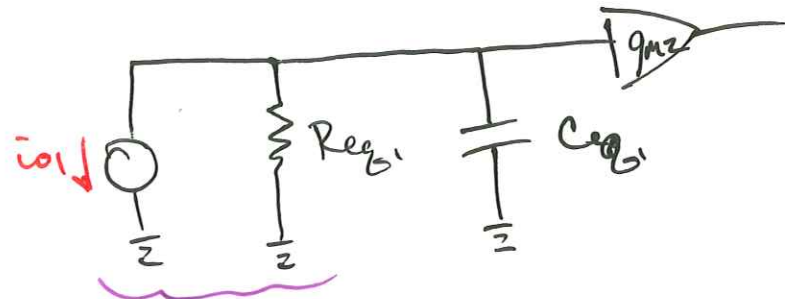
$$C_{eq1} = C_c(1 + g_{m2})$$

$$C_{eq2} = C_c \left(\frac{1 + g_{m2}}{g_{m2}} \right)$$

if g_{m2} is ideal, then
 $r_o \rightarrow 0 \rightarrow C_{eq2}$ is insignificant.

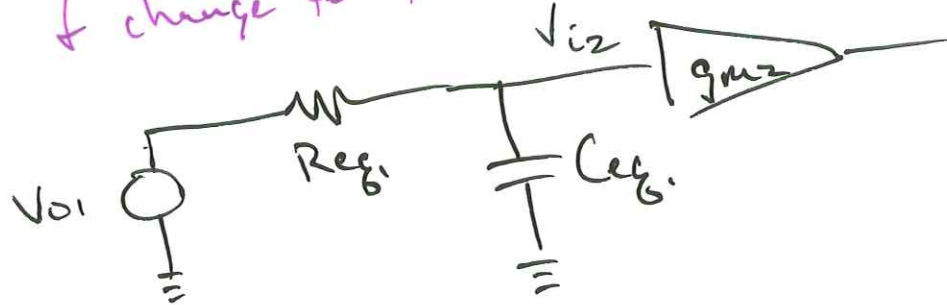


input stage is a non-ideal current output diff-amp:



Norton Eq. Source

↓ change to Thevenin



$$\frac{V_{i2}}{V_{o1}} = \frac{1}{1 + j2\pi f R_{eq1} C_{eq1}} = \frac{1}{1 + s R_{eq1} C_{eq1}} = \frac{1}{1 + s/f_p}$$

$$\therefore \text{pole } f: f_p = \frac{1}{2\pi R_{eq1} C_{eq1}}$$

$$741: C_c = 30 \text{ pF}, R_{eq1} = 1.95 \text{ M}\Omega, g_{m2} = 544$$

$$\rightarrow C_{eq1} = 16.3 \text{ nF}$$

$$\rightarrow f_p = 5 \text{ Hz}$$

Dominant Pole Amplifier Model

Low f pole dominates operation of Amp:

$$a(jf) = \frac{a_0}{1 + jf/f_b}$$

f_b = Open Loop Bandwidth
 $\rightarrow$ low-freq pole freq

f_t = transition frequency
 $\rightarrow |a(jf_t)| = 0\text{dB}$

$$0\text{dB} = 1 = \left| \frac{a_0}{1 + jf_t/f_b} \right| = \frac{a_0}{\sqrt{1 + (f_t/f_b)^2}} \quad \frac{f_t}{f_b} \gg 1$$

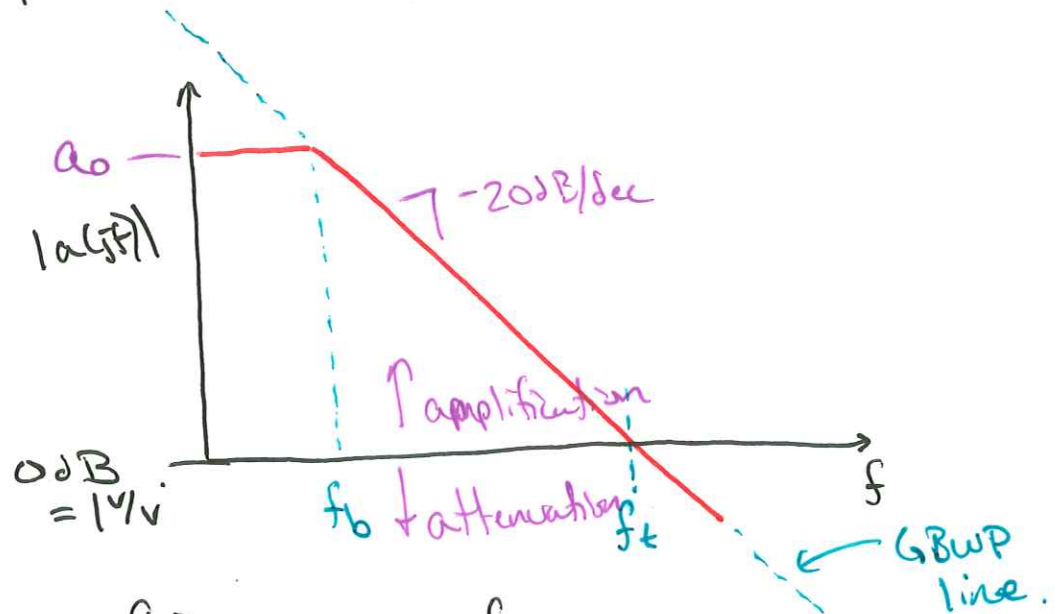
$$1 \approx \frac{a_0}{f_t/f_b} \rightarrow \underline{f_t} \approx \underline{a_0 f_b} \rightarrow \text{GBWP}$$

GBWP = Gain Bandwidth Product

$\rightarrow$ GBWP allows for estimation of any $a(jf)$ ~~from~~ at any freq.

$$|a(jf)| \approx a_0 / f/f_b \rightarrow |a(jf)| f \approx a_0 f_b = f_t$$

$$\rightarrow |a(jf)| \approx \frac{\text{GBWP}}{f}$$



* Closed Loop Gain

$$A = \frac{A_{ideal}}{1 + 1/T} + \frac{a_{ft}}{1 + T}$$

fixed after lecture

$$\rightarrow A(jf) = \frac{A_{ideal}}{1 + 1/T(jf)} + \frac{a_{ft}(jf)}{1 + T(jf)}$$

assume a_{ft} very small. --

$$A(jf) \approx A_{ideal} / (1 + 1/T(jf))$$

$$A(jf) = A_{ideal} / \left(1 + \frac{1}{\beta} \frac{1 + jf/f_b}{a_o} \right)$$

want: $A(jf) \approx A_o / (1 + jf/f_B)$ $A_o = A_{ideal} / (1 + a_o \beta)$

$$\rightarrow f_B = f_b (1 + \underbrace{a_o \beta}_{\text{DC loop gain, typically } \gg 1}) \approx f_b a_o \beta = f_t \beta \rightarrow f_B \left(\frac{1}{\beta} \right) = f_t$$

DC loop gain,
typically $\gg 1$

for low pass filter - type operation,

~~$A_o = A_{ideal}$~~ . see below

if completely resistive.

also $a_{ft}(jf) \rightarrow a_{ft}$.

for resistive feedback,

$T(jf)$ only freq. dependant term.

$$T = -\frac{a v_d}{v_T} \rightarrow T(jf) = -a(jf) \frac{v_d}{v_T}$$

$$T(jf) = a(jf) \beta(jf) \quad \beta(jf) = -\frac{v_d}{v_T}$$

where β = "feedback factor"

β not necessarily b , but can be in some situations (ex. non-inv. Amp)

$\frac{1}{\beta}$ similar to a_o
for closed-loop response

Exam 1 study focus:

- Ideal analysis, Applying virtual short to get V_o/V_i
- Exact Analysis ~~at~~ $a \neq \infty$
 - 4050 (usually) $r_d \rightarrow \infty$, $r_o \rightarrow 0$
 - 6050 (usually) one or both not ideal.

$$V_o = a(V_p - V_n)$$

- Return Ratio Analysis → calculate $T = -\frac{aV_d}{V_T}$
- know difference between a & a_{∞}
- Calculate input/output Resistances (and others)
 - Directly or via Blackman's formula.
- know when ~~the~~ terms can be dropped (for example:
 $\frac{1}{r_d} + \frac{1}{r_o} \approx \frac{1}{r_o}$)

-
- Diff Amps, CMRR
 - ... Instrumentation Amps

-
- Power Consumption / current flow /
 - Saturation / Voltage Compliance ...

→ $V-V_i$, $V-I$, $I-V$, $I-I$ $A = A_{ideal} [\tilde{v}_i \text{ or } \tilde{i}_i] - \frac{V_L}{R_o}$